Virtual Memory

Computer Architecture

Readings

- *Digital Design and Computer Architecture – David Harris & Sarah Harris*
  - Chapter 8

Virtual Address Translation

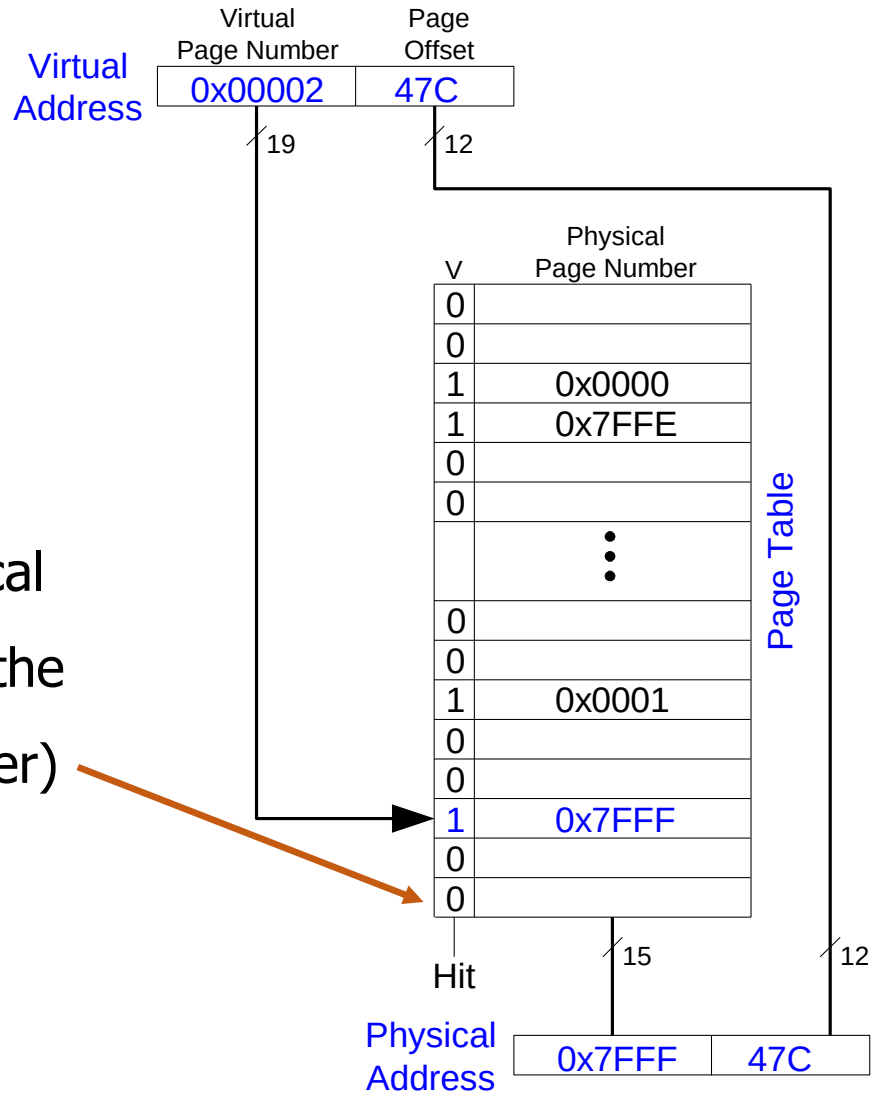
How Do We Translate Addresses?

- **Page table**
  - Has entry for each virtual page

- Each page table entry has:
  - Valid bit: whether the virtual page is located in physical memory (if not, it must be fetched from the hard disk)
  - Physical page number: where the virtual page is located in physical memory
  - (Replacement policy, dirty bits)

Page Table Example

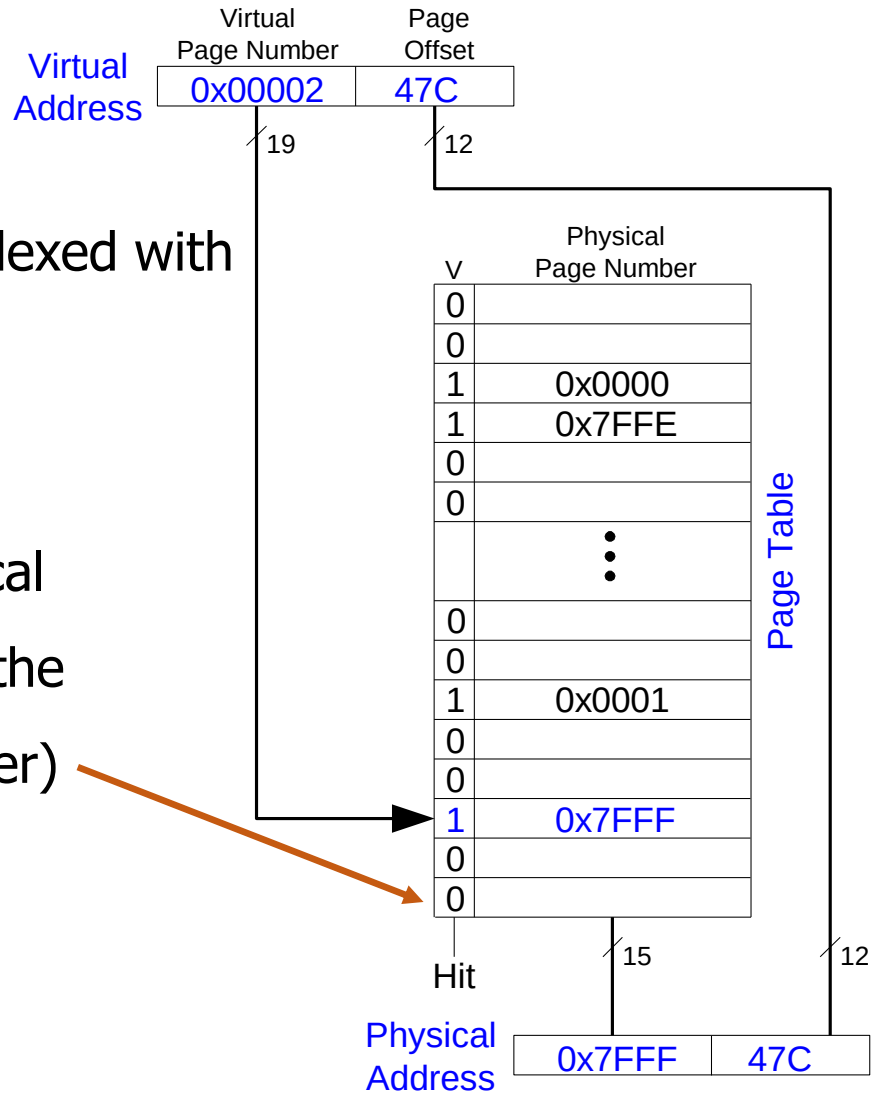
Page Table is located at physical memory address specified by the PTBR (Page Table Base Register)



Page Table Example

Page Table is Indexed with the VPN

Page Table is located at physical memory address specified by the PTBR (Page Table Base Register)

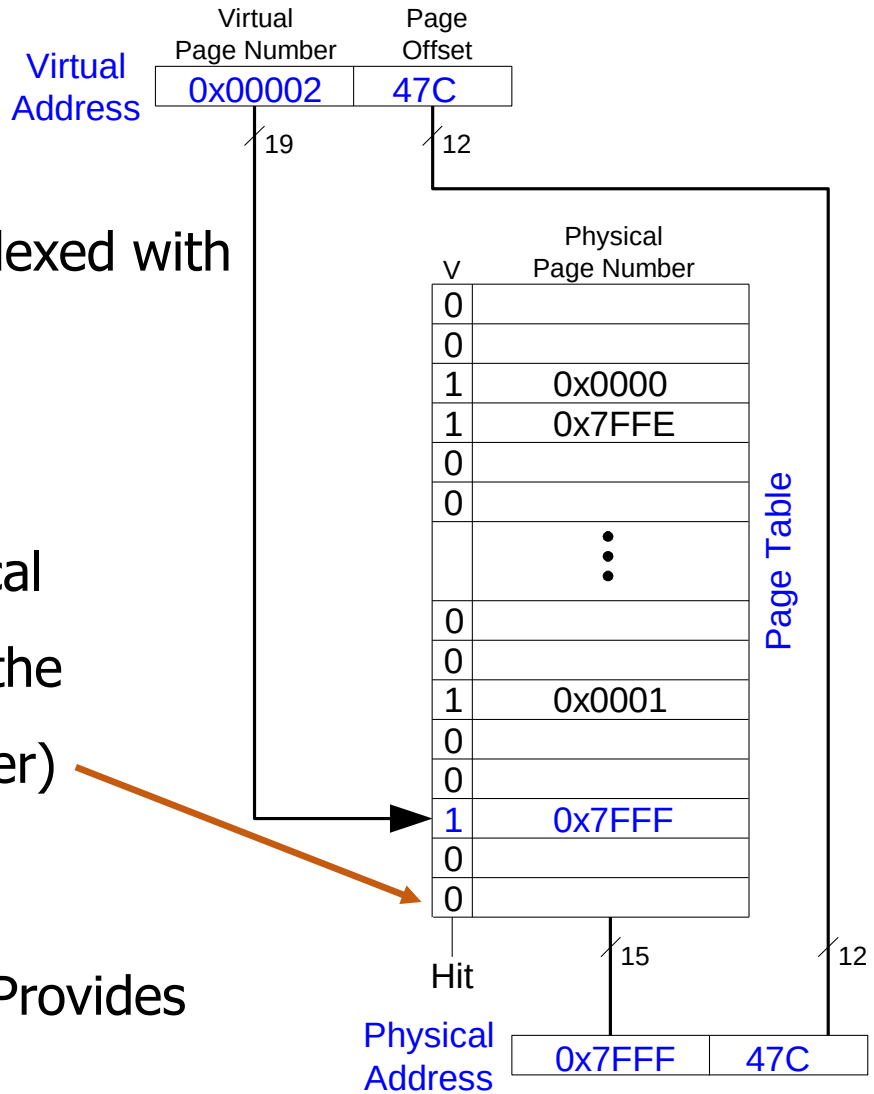


Page Table Example

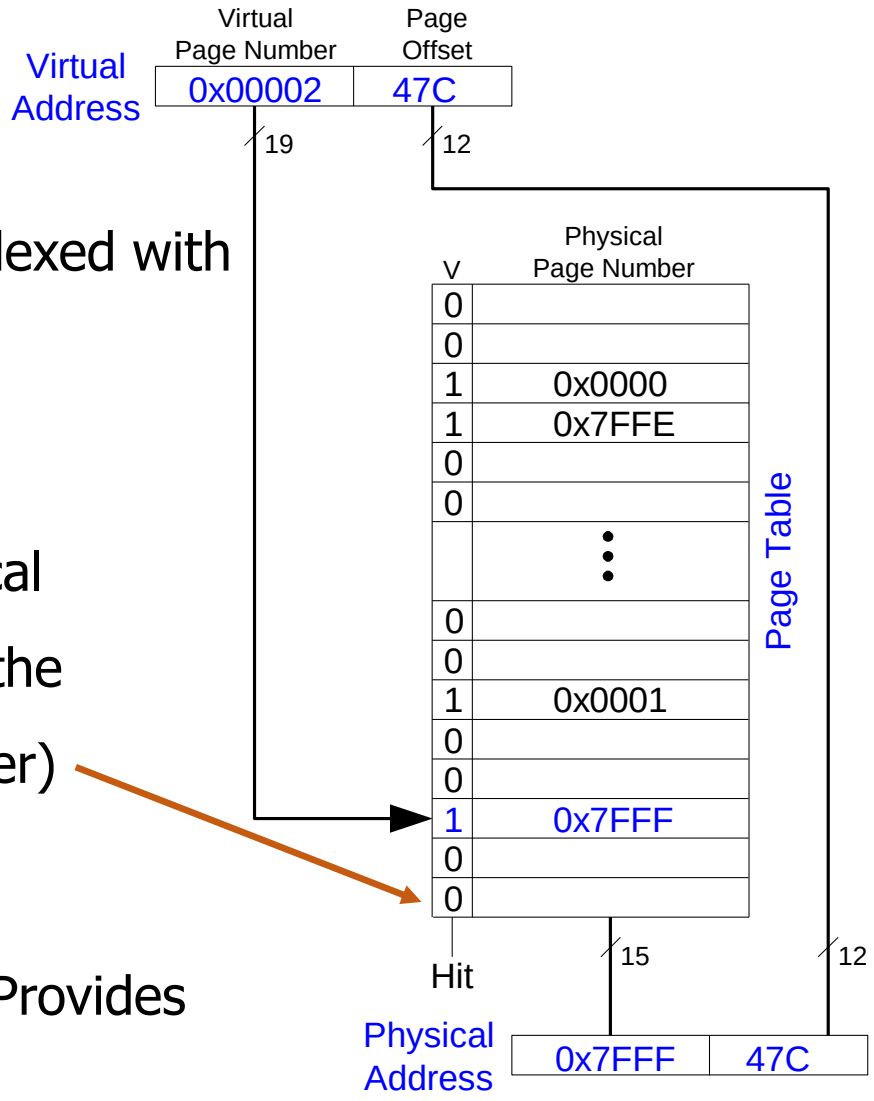
Page Table is Indexed with the VPN

Page Table is located at physical memory address specified by the PTBR (Page Table Base Register)

Page Table Provides the PPN



Page Table Example



Page Table is Indexed with the VPN

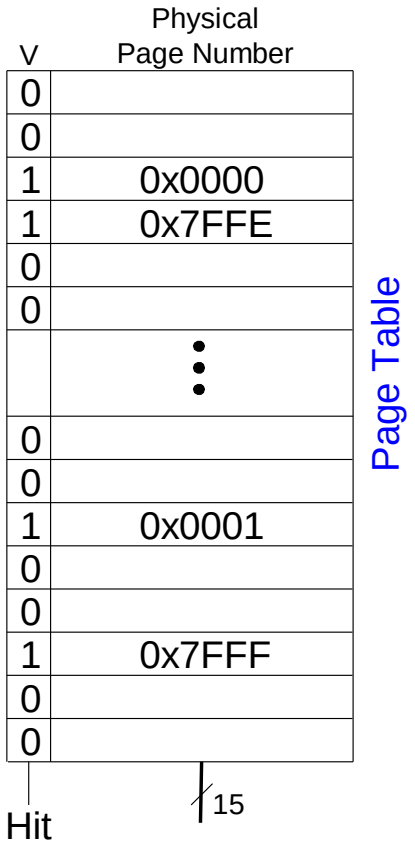
Page Table is located at physical memory address specified by the PTBR (Page Table Base Register)

Page Table Provides the PPN

Page Offset Bits do not change during Translation

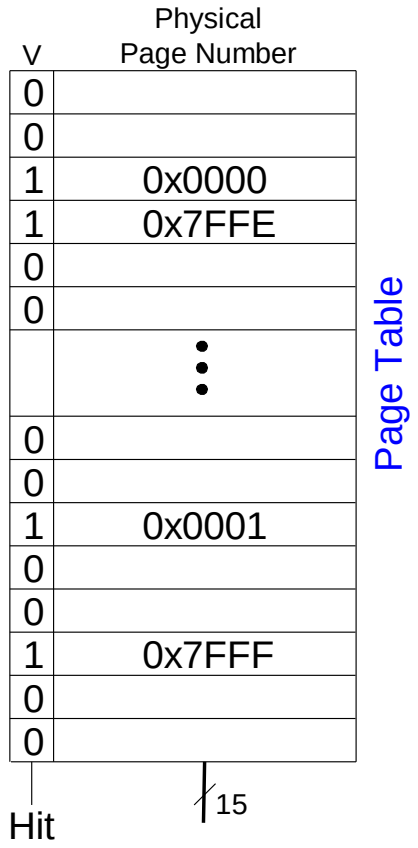
Page Table Example 2

- What is the physical address of virtual address 0x5F20?



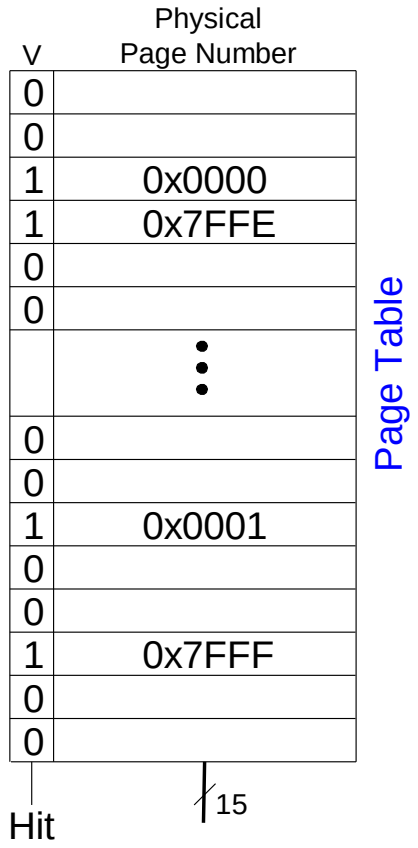
Page Table Example 2

- What is the physical address of virtual address 0x5F20?
- We first need to find the page table entry containing the translation for the corresponding VPN



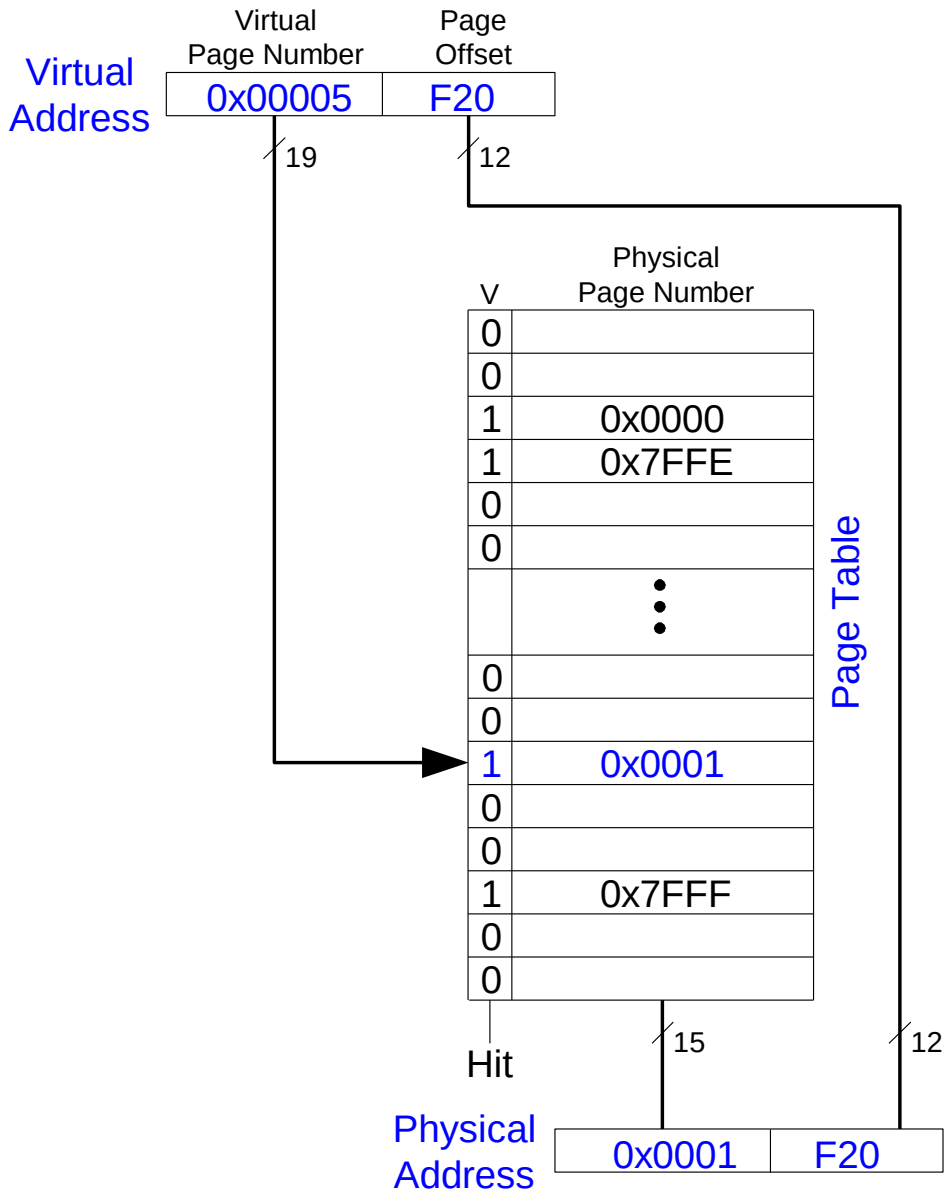
Page Table Example 2

- What is the physical address of virtual address 0x5F20?
- We first need to find the page table entry containing the translation for the corresponding VPN
- Look up the PTE at the address
  - PTBR + VPN * PTE-size



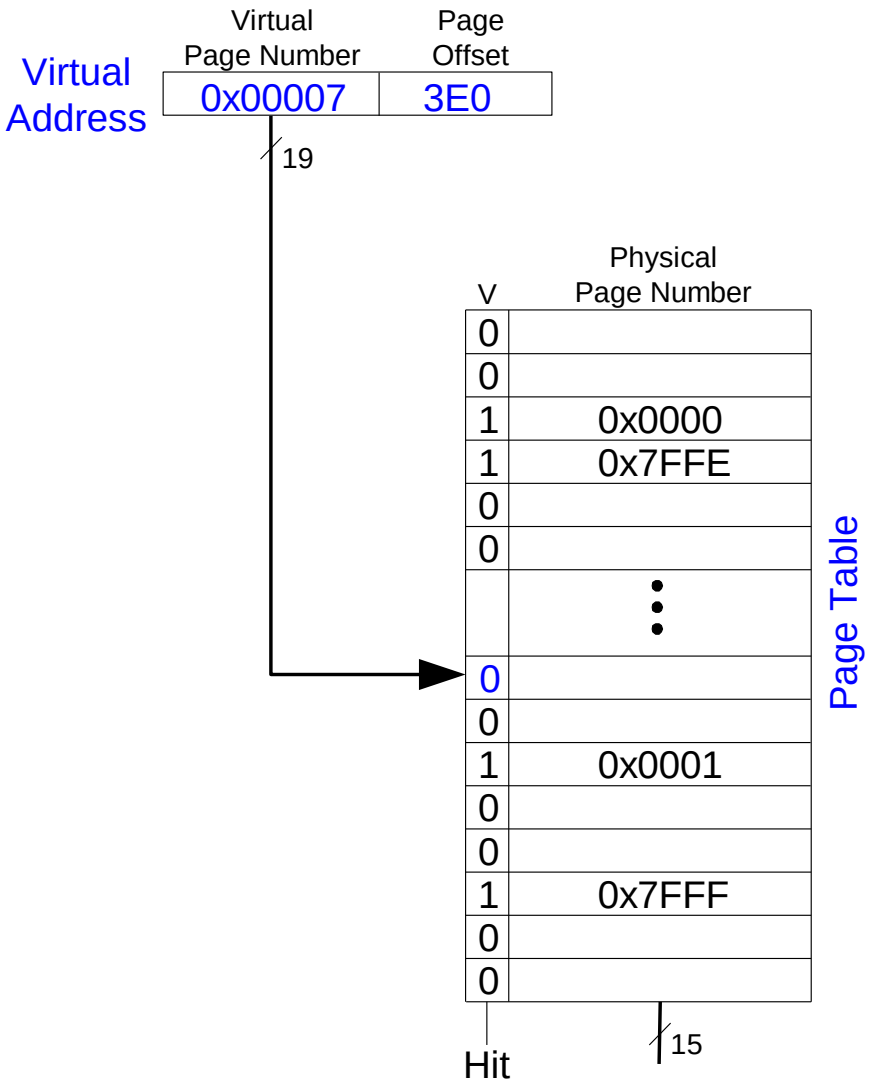
Page Table Example 2

- What is the physical address of virtual address 0x5F20?
  - VPN = 5
  - Entry 5 in page table indicates VPN 5 is in physical page 1
  - Physical address is 0x1F20

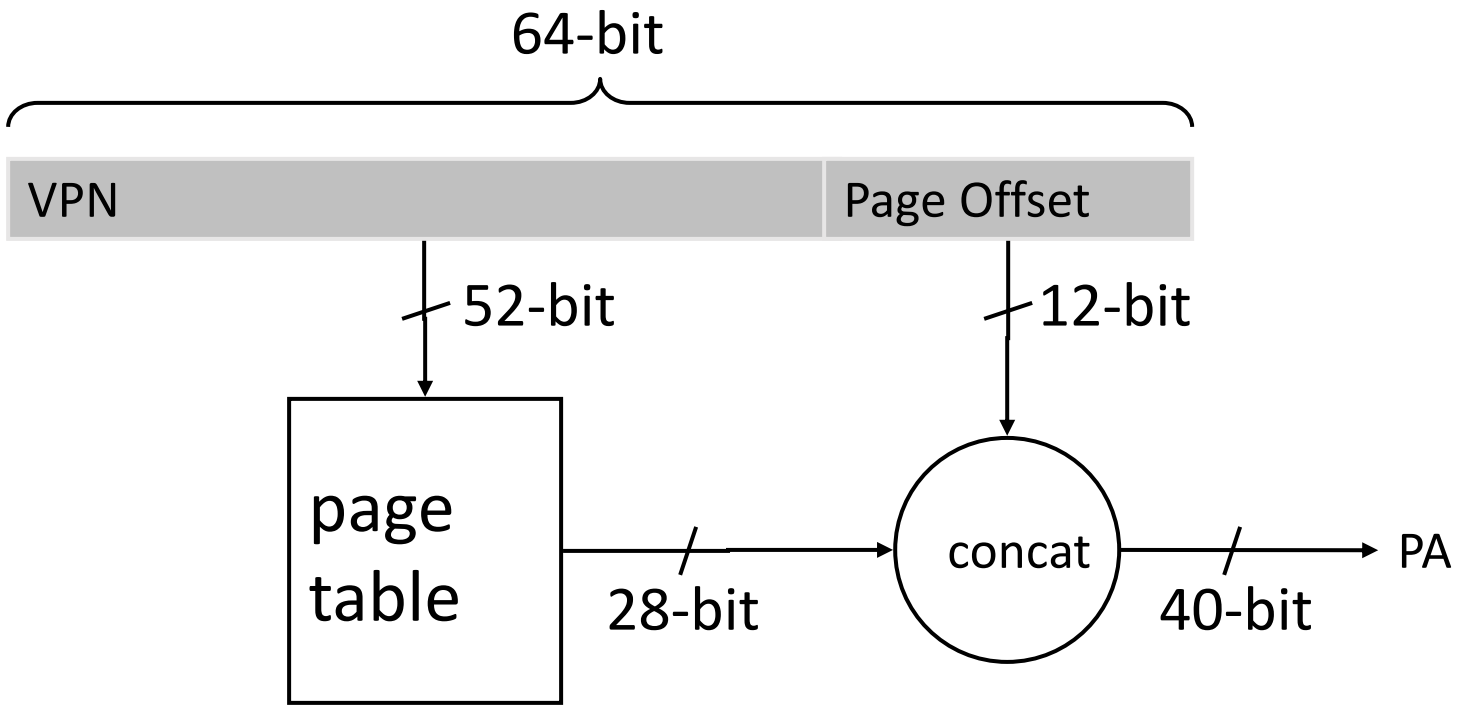


Page Table Example 3

- What is the physical address of virtual address 0x73E0?
  - VPN = 7
  - Entry 7 in page table is invalid, so the page is not in physical memory
  - The virtual page must be swapped into physical memory from disk



Issue: Page Table Size



- Suppose 64-bit VA and 40-bit PA, how large is the page table?
  - **2^{52} entries x ~4 bytes ≈ 2^{54} bytes**

and that is for just one process!

and the process may not be using the entire VM space!

Page Table Challenges

- Challenge 1: Page table is large
  - at least part of it needs to be located in physical memory
  - Solution: multi-level (hierarchical) page tables

- Challege 2: Each load/store requires at least two memory accesses:
  1. one for address translation (page table read)
  2. one to access data with the physical address (after translation)

- Two memory accesses to service a load/store greatly degrades load/store execution time
  - Unless we are clever… -> speed up the translation

Translation Lookaside Buffer (TLB)

- Idea: Cache the page table entries (PTEs) in a hardware structure in the processor

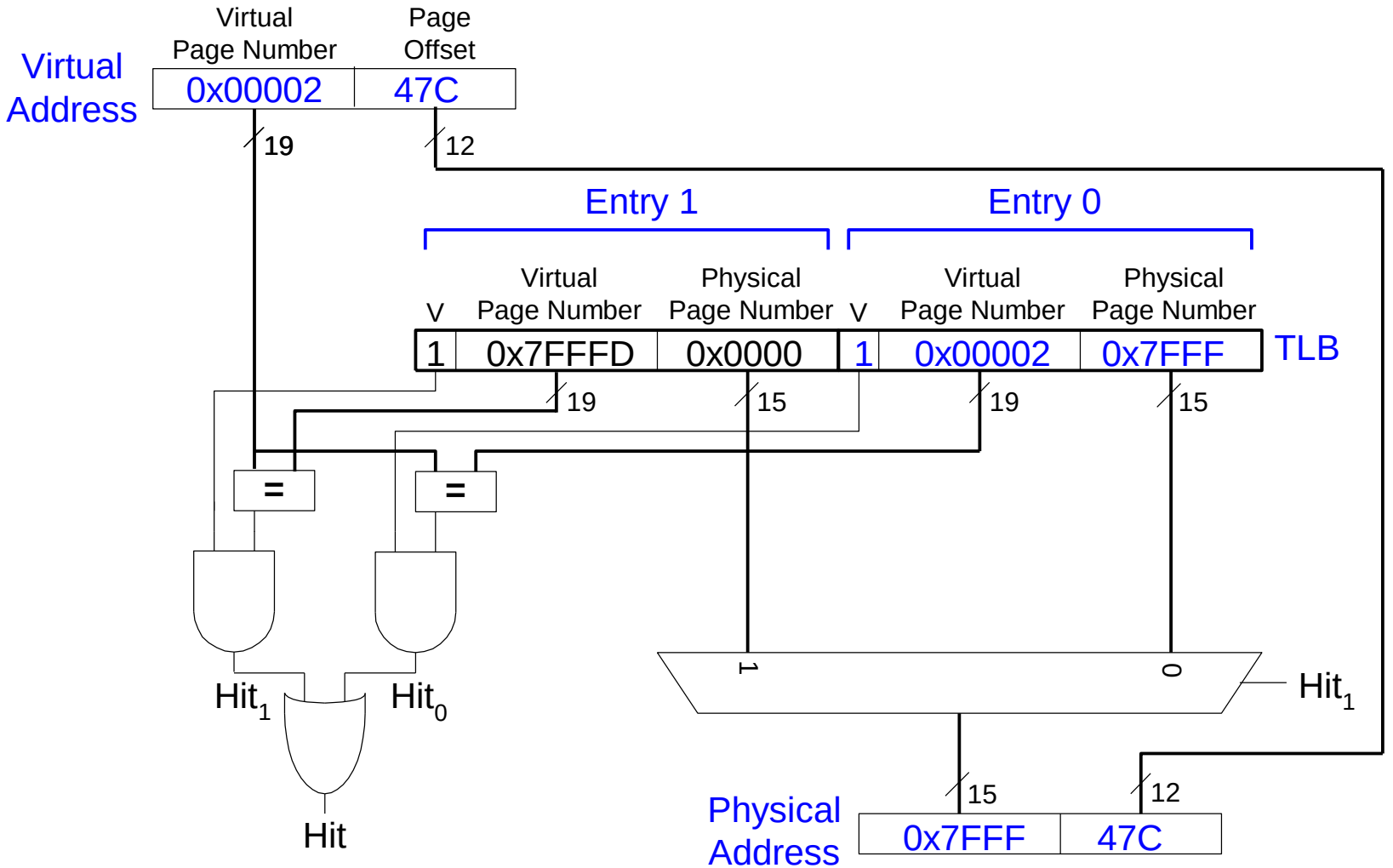
- Translation lookaside buffer (TLB)
  - Small cache of most recently used translations (PTEs)
  - Reduces number of memory accesses required for *most* loads/stores to only one

Translation Lookaside Buffer (TLB)

- Page table accesses have a lot of temporal locality
  - Data accesses have temporal and spatial locality
  - Large page size (say 4KB, 8KB, or even 1-2GB), so consecutive loads/stores likely to access same page

- TLB
  - Small: accessed in ~ 1 cycle
  - Typically 16 - 512 entries
  - High associativity
  - > 95-99 % hit rates typical (depends on workload)
  - Reduces # of memory accesses for most loads and stores to only 1

Example Two-Entry TLB



Virtual Memory Support and Examples

Supporting Virtual Memory

- Virtual memory requires both HW+SW support
  - Page Table is in memory
  - Can be cached in special hardware structures called Translation Lookaside Buffers (TLBs)
- The hardware component is called the MMU (memory management unit)
  - Includes Page Table Base Register(s), TLBs, page walkers
- It is the job of the software to leverage the MMU to
  - Populate page tables, decide what to replace in physical memory
  - Change the Page Table Register on context switch (to use the running thread's page table)
  - Handle page faults and ensure correct mapping

Address Translation

- How to obtain the physical address from a virtual address?

- Page size specified by the ISA
  - VAX: 512 bytes
  - Today: 4KB, 8KB, 2GB, … (small and large pages mixed together)
  - Trade-offs? (remember cache lectures)

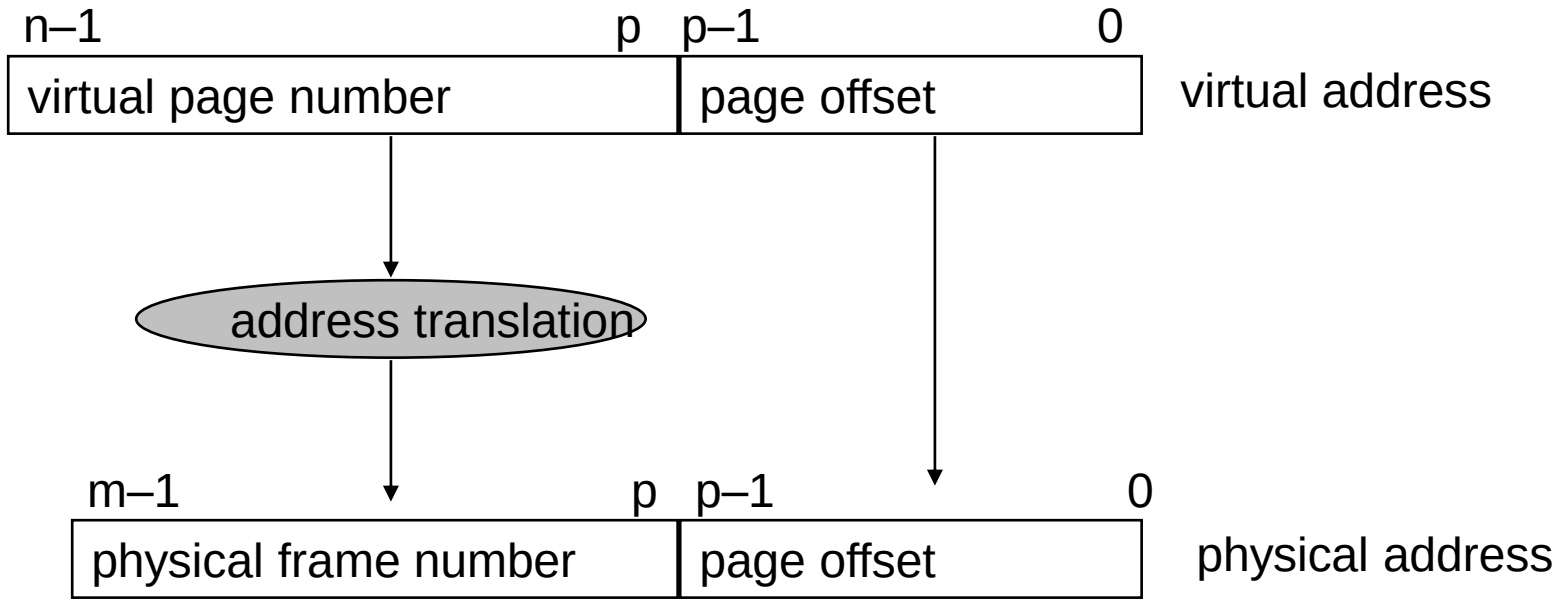
- Page Table contains an entry for each virtual page
  - Called Page Table Entry (PTE)
  - What is in a PTE?

What Is in a Page Table Entry (PTE)?

- Page table is the "tag store" for the physical memory data store
  - A mapping table between virtual memory and physical memory
- PTE is the "tag store entry" for a virtual page in memory
  - Need a valid bit → to indicate validity/presence in physical memory
  - Need tag bits (PFN) → to support translation
  - Need bits to support replacement
  - Need a dirty bit to support "write back caching"
  - Need protection bits to enable access control and protection

Address Translation (I)

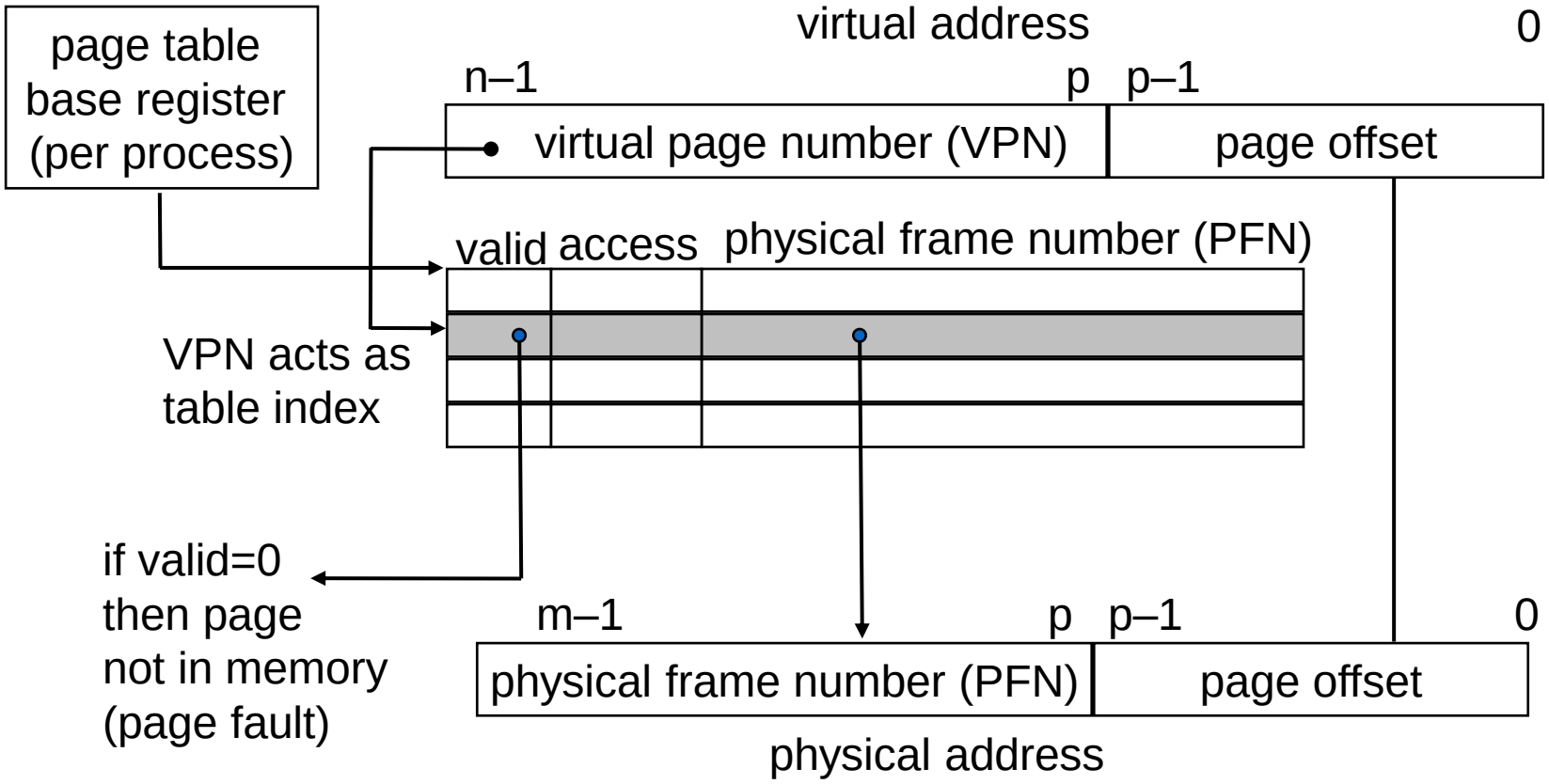
- Parameters
  - P = 2^p = page size (bytes).
  - N = 2^n = Virtual-address limit
  - M = 2^m = Physical-address limit



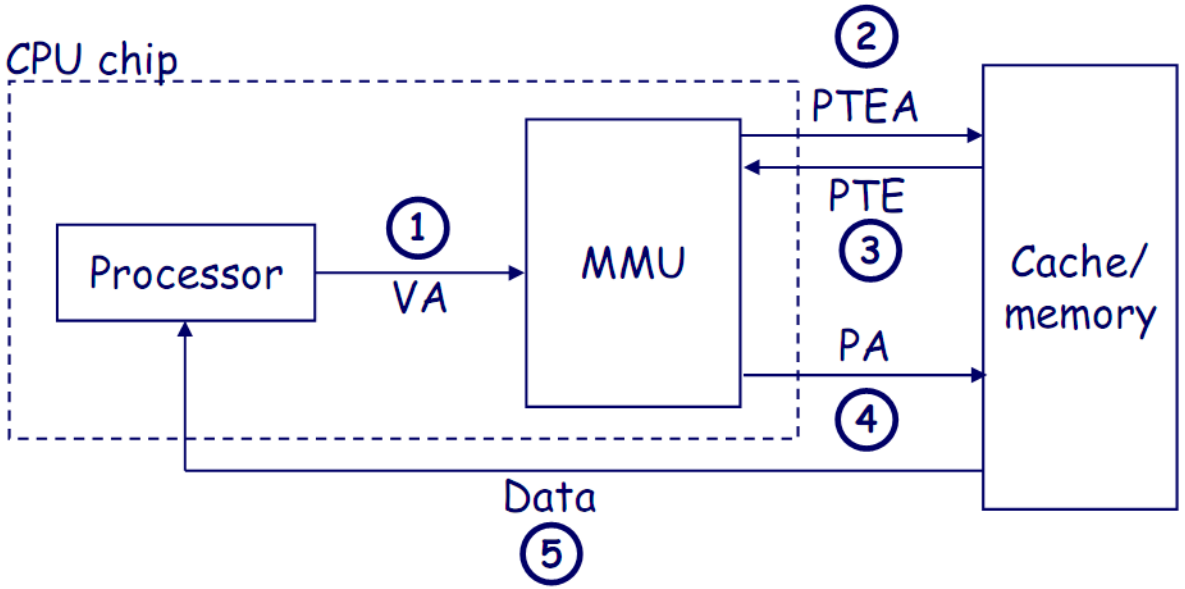
Page offset bits don't change as a result of translation

Address Translation (II)

- Separate (set of) page table(s) per process
- VPN forms index into page table (points to a page table entry)
- Page Table Entry (PTE) provides information about page



Address Translation: Page Hit



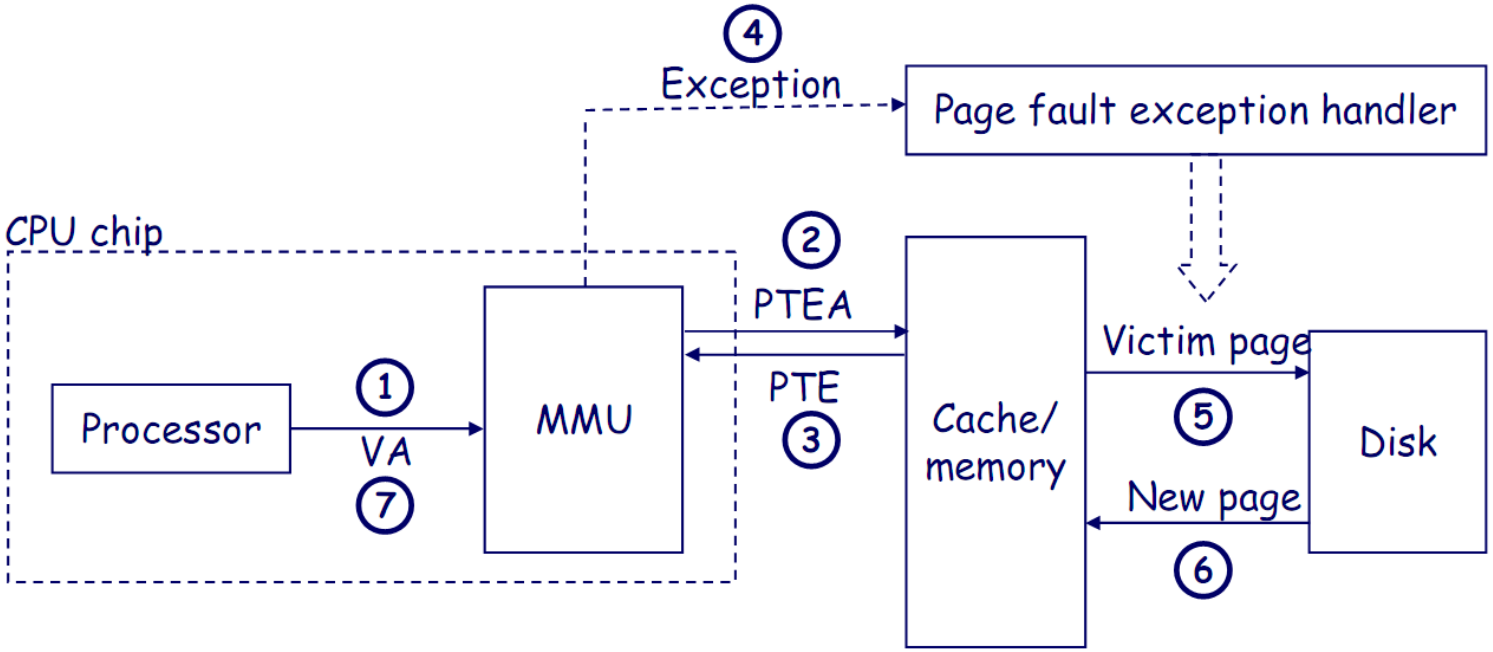
1) Processor sends virtual address to MMU

2-3) MMU fetches PTE from page table in memory

4) MMU sends physical address to L1 cache

5) L1 cache sends data word to processor

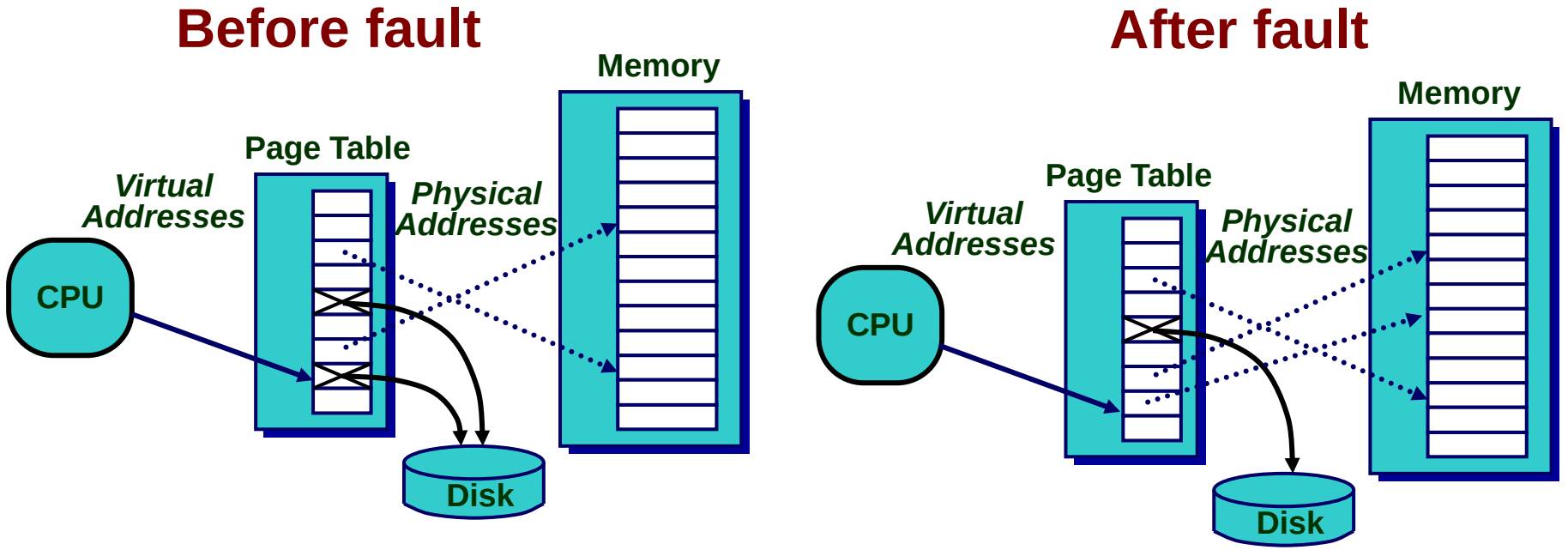
Address Translation: Page Fault



1) Processor sends virtual address to MMU
2-3) MMU fetches PTE from page table in memory
4) Valid bit is zero, so MMU triggers page fault exception
5) Handler identifies victim, and if dirty pages it out to disk
6) Handler pages in new page and updates PTE in memory
7) Handler returns to original process, restarting faulting instruction.

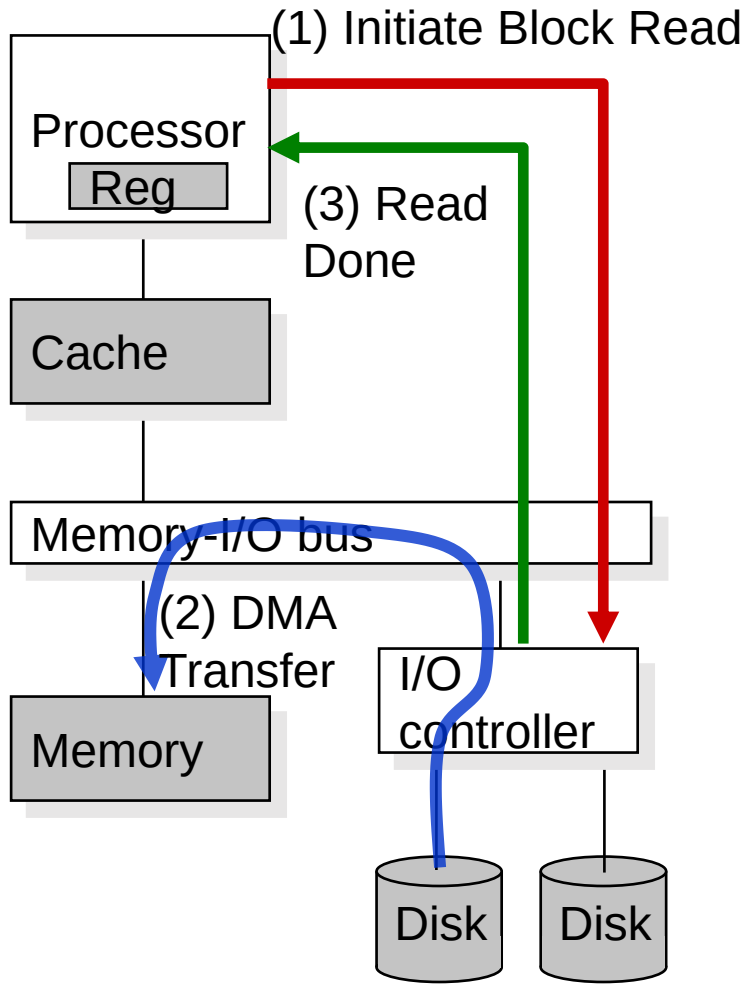
Page Fault ("A Miss in Physical Memory")

- If a page is not in physical memory but disk
  - Page table entry indicates virtual page not in memory
  - Access to such a page triggers a page fault exception
  - OS trap handler invoked to move data from disk into memory
    - Other processes can continue executing
    - OS has full control over placement



Servicing a Page Fault

- (1) Processor signals controller
  - Read block of length P starting at disk address X and store starting at memory address Y
- (2) Read occurs
  - Direct Memory Access (DMA)
  - Under control of I/O controller
- (3) Controller signals completion
  - Interrupt processor
  - OS resumes suspended process



Page Replacement Algorithms

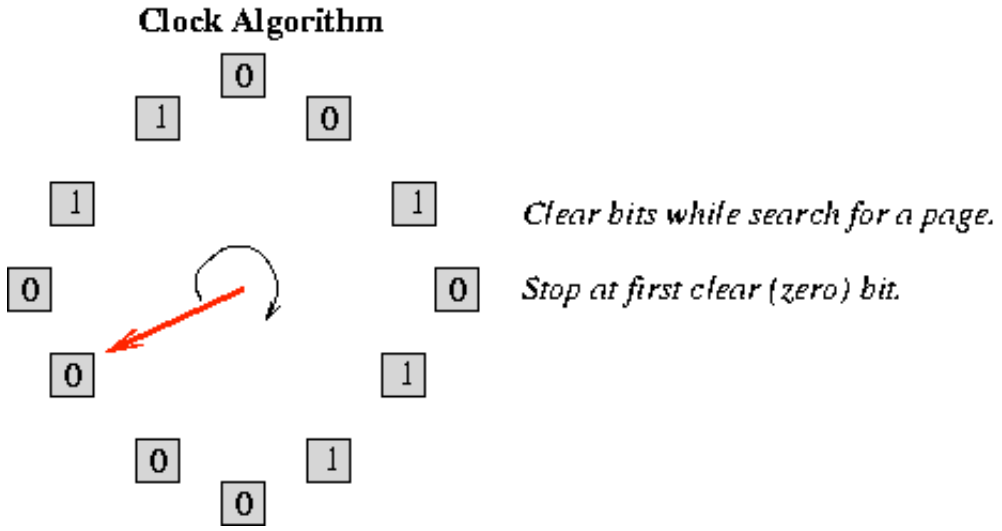
- If physical memory is full (i.e., list of free physical pages is empty), which physical frame to replace on a page fault?

- Is True LRU feasible?
  - 4GB memory, 4KB pages, how many possibilities of ordering?

- Modern systems use approximations of LRU
  - E.g., the CLOCK algorithm
- And, more sophisticated algorithms to take into account "frequency" of use
  - E.g., the ARC algorithm
  - Megiddo and Modha, "ARC: A Self-Tuning, Low Overhead Replacement Cache," FAST 2003.

CLOCK Page Replacement Algorithm

- Keep a circular list of physical frames in memory
- Keep a pointer (hand) to the last-examined frame in the list
- When a page is accessed, set the R bit in the PTE
- When a frame needs to be replaced, replace the first frame that has the reference (R) bit not set, traversing the circular list starting from the pointer (hand) clockwise
  - During traversal, clear the R bits of examined frames
  - Set the hand pointer to the next frame in the list



Memory Protection

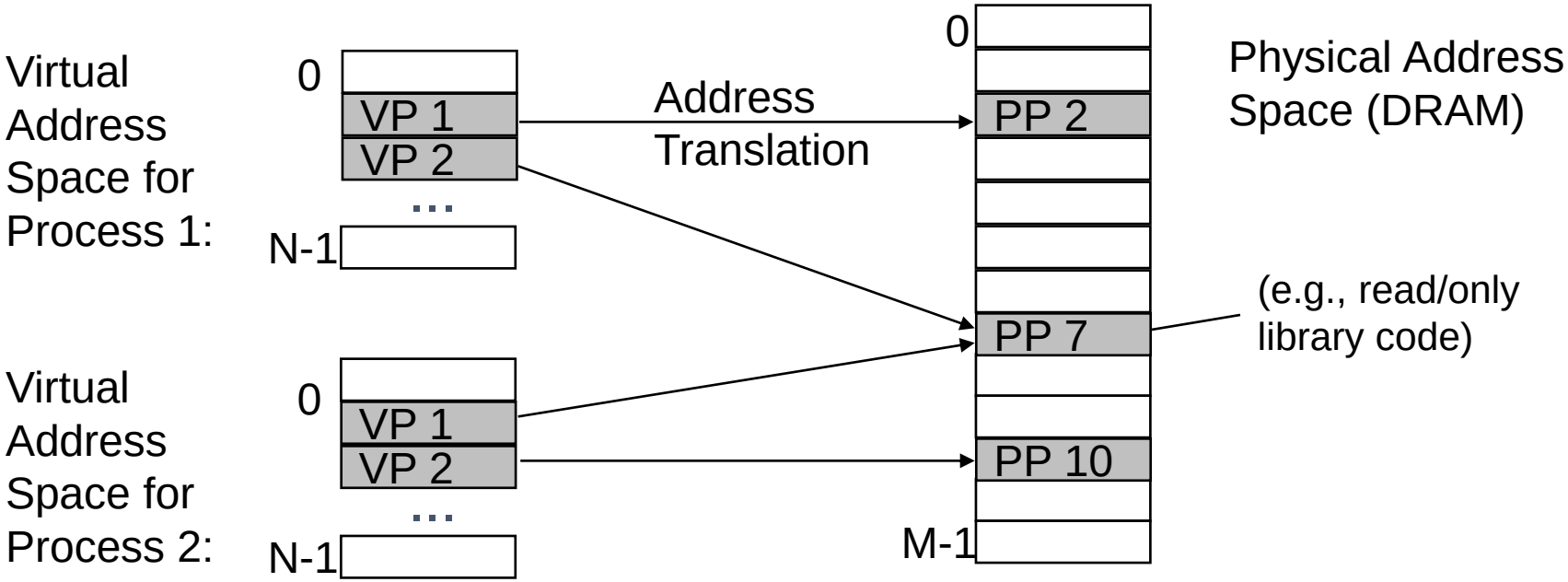
Memory Protection

- Multiple programs (*processes*) run at once
  - Each process has its own page table
  - Each process can use entire virtual address space without worrying about where other programs are

- A process can only access physical pages mapped in its page table – cannot overwrite memory of another process
  - Provides protection and isolation between processes
  - Enables access control mechanisms per page

Page Table is Per Process

- Each process has its own virtual address space
  - Full address space for each program
  - Simplifies memory allocation, sharing, linking and loading.



Access Protection/Control via Virtual Memory

Page-Level Access Control (Protection)

- Not every process is allowed to access every page
  - E.g., may need supervisor level privilege to access system pages

- Idea: Store access control information on a page basis in the process's page table

- Enforce access control at the same time as translation

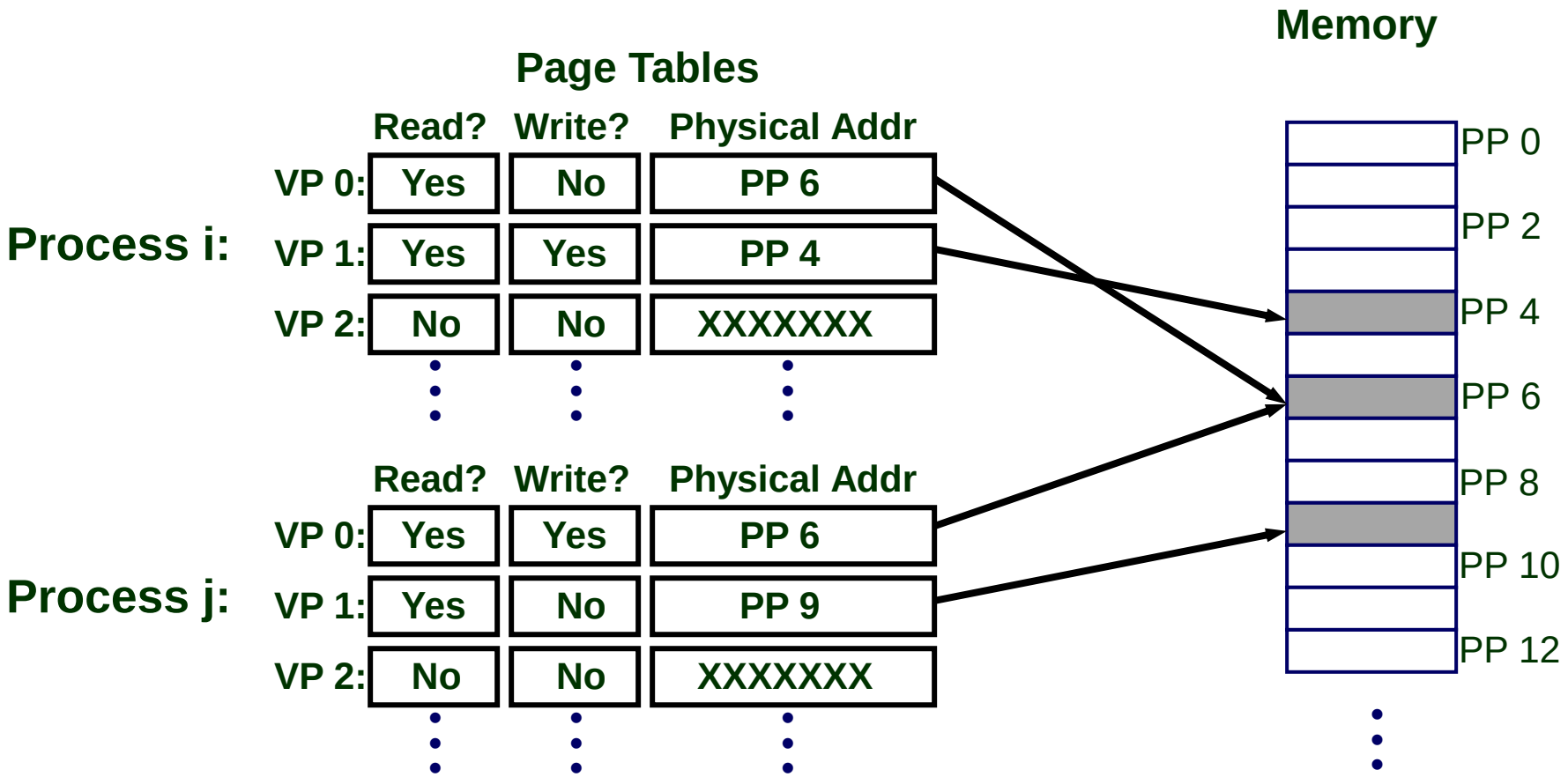
→ Virtual memory system serves two functions today

Address translation (for illusion of large physical memory)

Access control (protection)

VM as a Tool for Memory Access Protection

- Extend Page Table Entries (PTEs) with permission bits
- Check bits on each access and during a page fault
  - If violated, generate exception (Access Protection exception)



Virtual Memory Summary

- Virtual memory gives the illusion of "infinite" capacity
- A subset of virtual pages are located in physical memory
- A page table maps virtual pages to physical pages – this is called address translation
- A TLB speeds up address translation
- Using different page tables for different programs provides memory protection

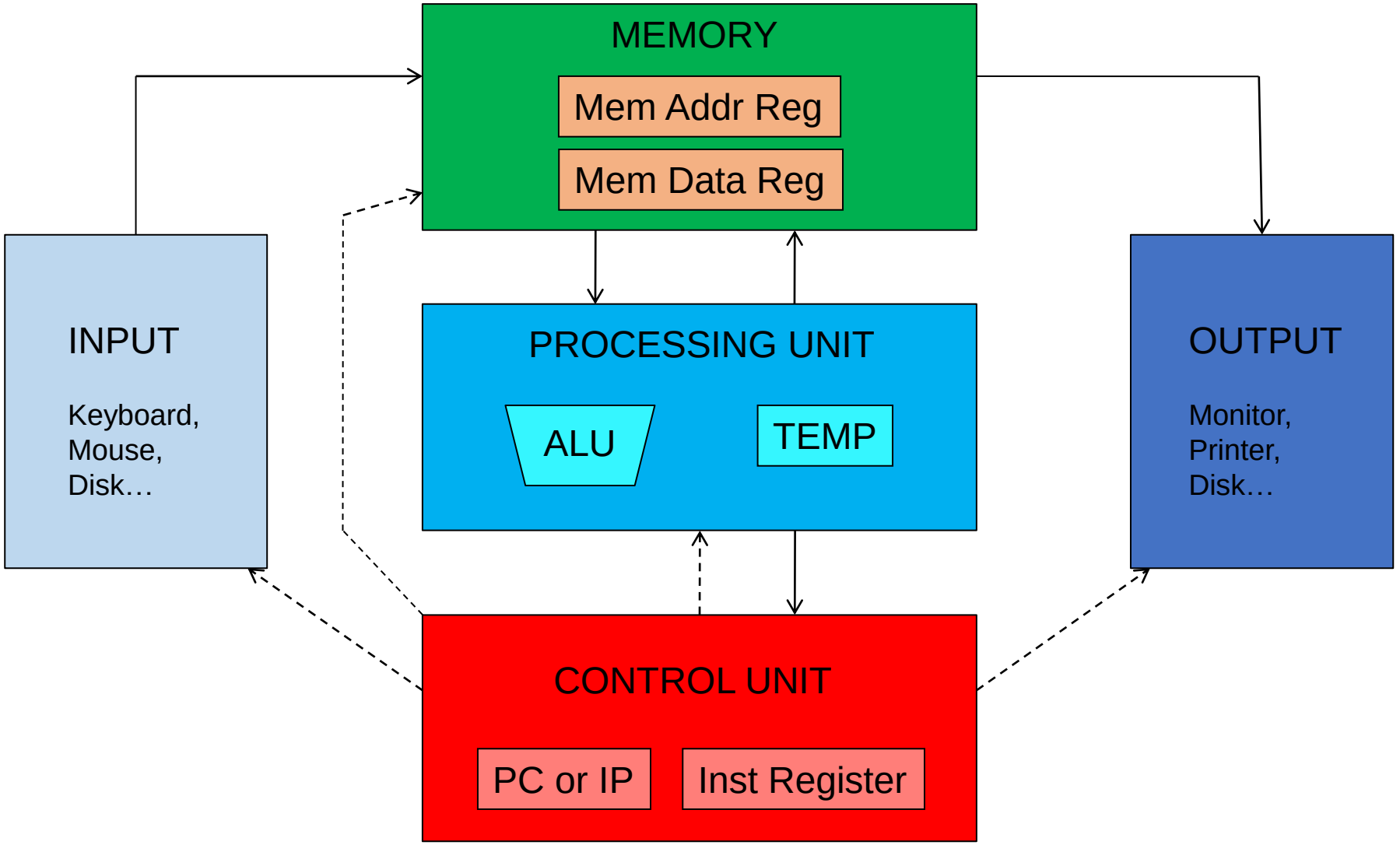
Microarchitecture

Readings

- Digital Design and Computer Architecture - Sarah Harris, David Harris
  - Chapter 7.1-7.3

- Instruction Set Architectures (ISA): LC-3
- Assembly programming: LC-3
- Memory Technologies, Memory Hierarchy
- Caches
- Virtual Memory
- Microarchitecture (principles & single-cycle uarch)
- Multi-cycle microarchitecture
- Pipelining
- Issues in Pipelining: Control & Data Dependence Handling, State Maintenance and Recovery, …
- …

Recall: The Von Neumann Model



Recall: LC-3: A Von Neumann Machine

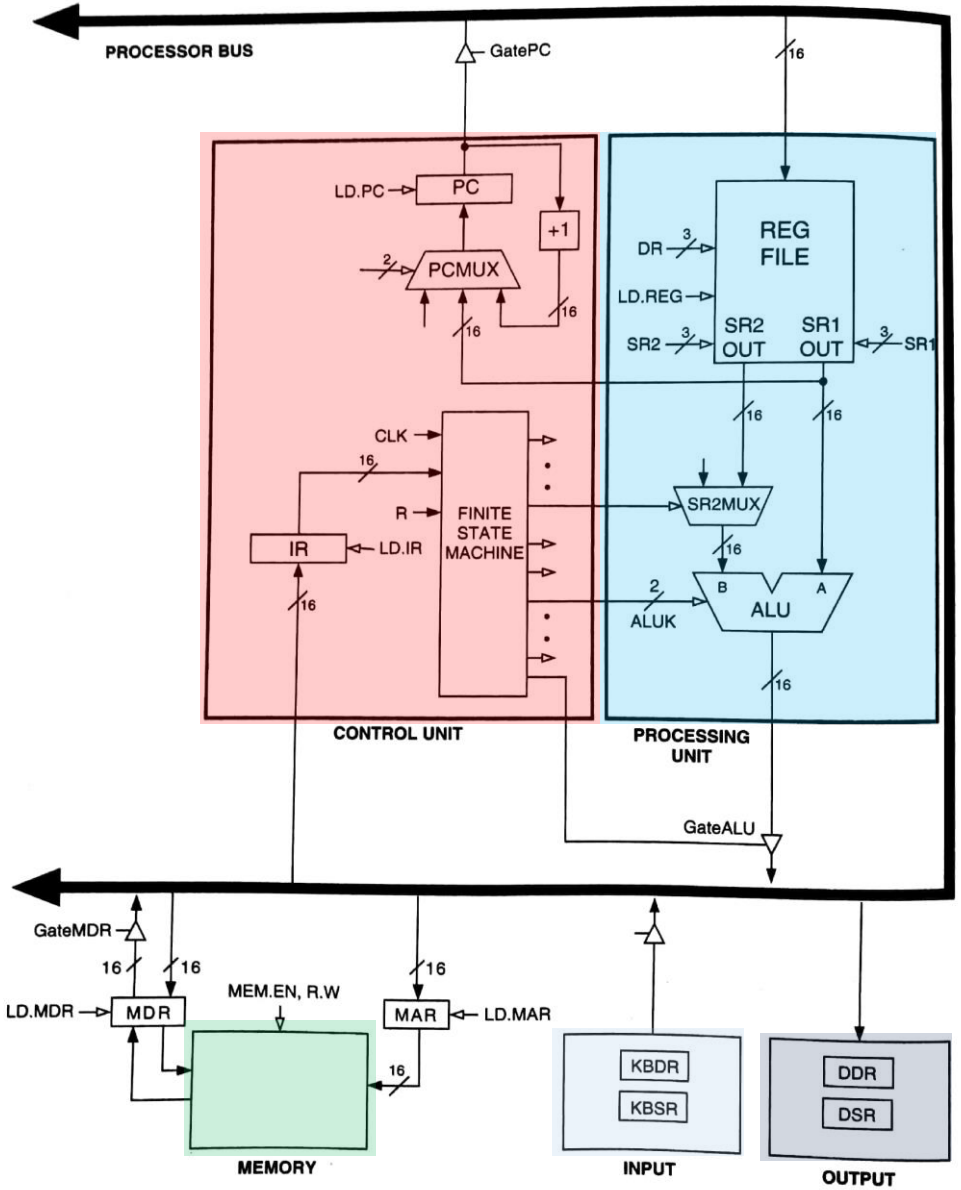
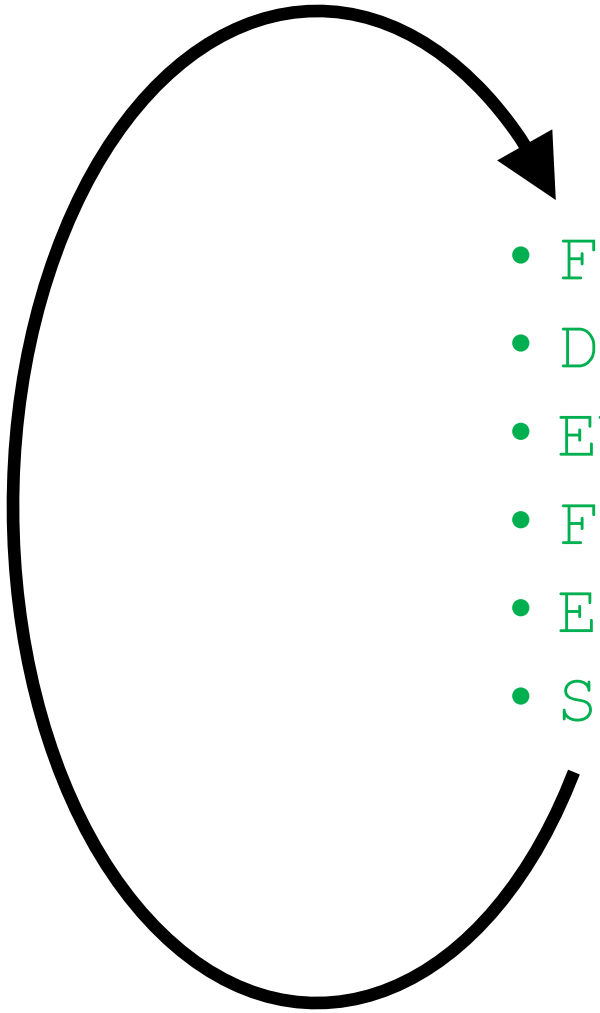


Figure 4.3 The LC-3 as an example of the von Neumann model

Recall: The Instruction Cycle

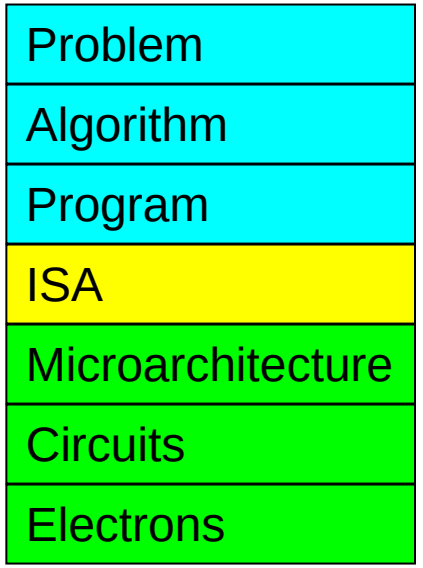


- FETCH
- DECODE
- EVALUATE ADDRESS
- FETCH OPERANDS
- EXECUTE
- STORE RESULT

Recall: The Instruction Set Architecture

- The ISA is the interface between what the software commands and what the hardware carries out

- The ISA specifies
  - The memory organization
    - Address space (LC-3: 2^{16}, MIPS: 2^{32})
    - Addressability (LC-3: 16 bits, MIPS: 32 bits)
    - Word- or Byte-addressable
  - The register set
    - R0 to R7 in LC-3
    - 32 registers in MIPS
  - The instruction set
    - Opcodes
    - Data types
    - Addressing modes
    - Semantics of instructions



Microarchitecture

- An **implementation** of the ISA

- How do we implement the ISA?
  - We will discuss this for many lectures

- There can be many implementations of the same ISA
  - MIPS R2000, R10000, …
  - Intel 80486, Pentium, Pentium Pro, Pentium 4, Kaby Lake, Coffee Lake, … AMD K5, K7, K9, Bulldozer, BobCat, …

The Von-Neumann Model

- All major *instruction set architectures* today use this model
  - x86, ARM, MIPS, SPARC, Alpha, POWER, RISC-V, …
- Underneath (at the microarchitecture level), the execution model of almost all *implementations (or, microarchitectures)* is very different
  - Pipelined instruction execution: *Intel 80486 uarch*
  - Multiple instructions at a time: *Intel Pentium uarch*
  - Out-of-order execution: *Intel Pentium Pro uarch*
  - Separate instruction and data caches
- But, what happens underneath that is ***not* consistent** with the von Neumann model is ***not* exposed** to software
  - Difference between ISA and microarchitecture

Property of ISA vs. Uarch?

- ADD instruction's opcode
- Bit-serial adder vs. Ripple-carry adder
- Number of general purpose registers
- Number of cycles to execute the MUL instruction
- Number of ports to the register file
- Whether or not the machine employs pipelined in struction execution

- Remember
  - Microarchitecture: Implementation of the ISA under sp ecific design constraints and goals

Now That We Have an ISA

- How do we implement it?
- i.e., how do we design a system that obeys the hardware/software interface?
- Aside: "System" can be solely hardware or a combination of hardware and software
  - "Translation of ISAs"
  - A **virtual ISA** can be converted by "software" into an **implementation ISA**
- We will assume "hardware" implementation for most lectures

How Does a Machine Process Instructions?

- What does processing an instruction mean?
- We will assume the von Neumann model (for now)

AS = Architectural (programmer visible) state before an instruction is processed

Process instruction

AS' = Architectural (programmer visible) state after an instruction is processed

- Processing an instruction: Transforming AS to AS' according to the ISA specification of the instruction

The Von Neumann Model/Architecture

Stored program

Sequential instruction processing

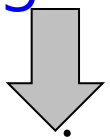
The "Process Instruction" Step

- ISA specifies abstractly what AS' should be, given an instruction and AS
  - It defines an abstract finite state machine where
    - State = programmer-visible state
    - Next-state logic = instruction execution specification
  - From ISA point of view, there are no "intermediate states" between AS and AS' during instruction execution
    - One state transition per instruction
- Microarchitecture implements how AS is transformed to AS'
  - There are many choices in implementation
  - We can have programmer-invisible state to optimize the speed of instruction execution: **multiple** state transitions per instruction
    - Choice 1: AS → AS' (transform AS to AS' in a single clock cycle)
    - Choice 2: AS → AS+MS1 → AS+MS2 → AS+MS3 → AS' (take multiple clock cycles to transform AS to AS')

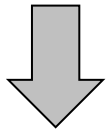
A Very Basic Instruction Processing Engine

- Each instruction takes a single clock cycle to execute
- Only combinational logic is used to implement instruction execution
  - *No intermediate, programmer-invisible state updates*

AS = Architectural (programmer visible) state
at the beginning of a clock cycle



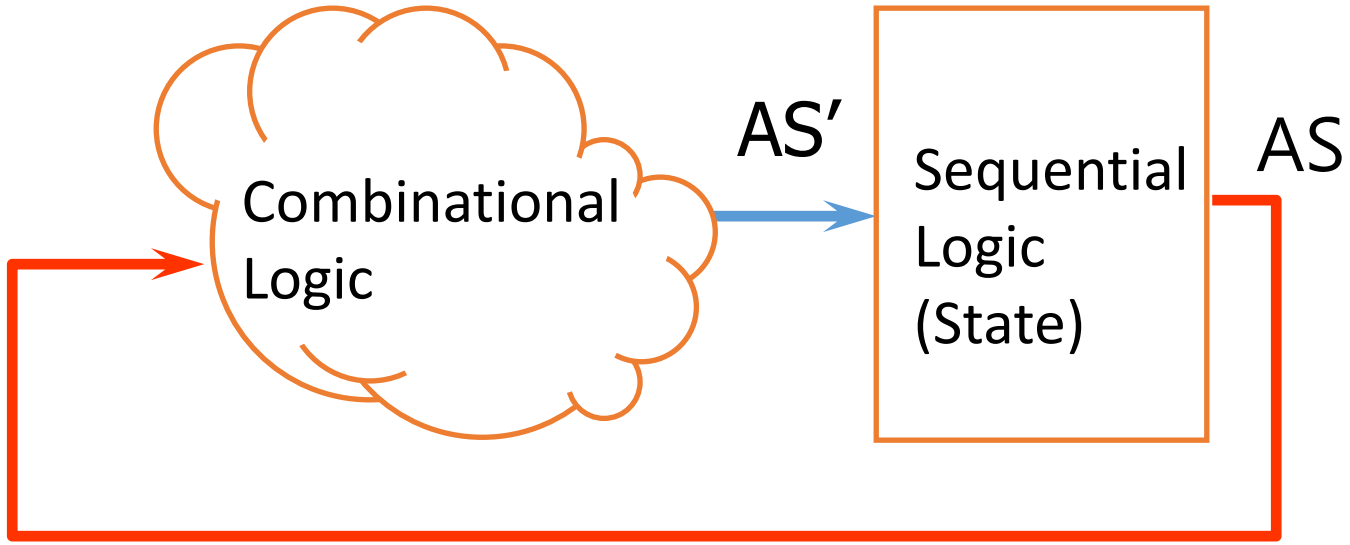
Process instruction in one clock cycle



AS' = Architectural (programmer visible) state
at the end of a clock cycle

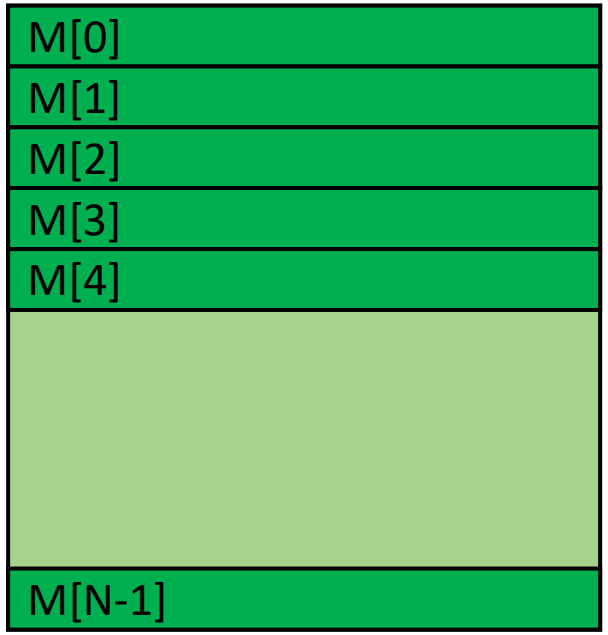
A Very Basic Instruction Processing Engine

- Single-cycle machine



- What is the *clock cycle time* determined by?
- What is the *critical path* of the combinational logic determined by?

Recall: Programmer Visible (Architectural) State



Memory
array of storage locations
indexed by an address

Registers
- given special names in the ISA
  (as opposed to addresses)
- general vs. special purpose

Program Counter
memory address
of the current instruction

Instructions (and programs) specify how to transform
the values of programmer visible state

Single-cycle vs. Multi-cycle Machines

- Single-cycle machines
  - Each instruction takes a single clock cycle
  - All state updates made at the end of an instruction's execution
  - Big disadvantage: The slowest instruction determines cycle time → long clock cycle time

- Multi-cycle machines
  - Instruction processing broken into multiple cycles/stages
  - State updates can be made during an instruction's execution
  - Architectural state updates made at the end of an instruction's execution
  - Advantage over single-cycle: The slowest "stage" determines cycle time

- Both single-cycle and multi-cycle machines literally follow the von Neumann model at the microarchitecture level

Instruction Processing “Cycle”

- Instructions are processed under the direction of a “control unit” step by step.
- Instruction cycle: Sequence of steps to process an instruction
- Fundamentally, there are six steps:

- Fetch
- Decode
- Evaluate Address
- Fetch Operands
- Execute
- Store Result

- Not all instructions require all six steps

Instruction Processing "Cycle" vs. Machine Clock Cycle

- Single-cycle machine:
  - All six phases of the instruction processing cycle take a *single machine clock cycle* to complete
- Multi-cycle machine:
  - All six phases of the instruction processing cycle can take *multiple machine clock cycles* to complete
  - In fact, each phase can take multiple clock cycles to complete

Instruction Processing Viewed Another Way

- Instructions transform Data (AS) to Data' (AS')
- This transformation is done by functional units
  - Units that "operate" on data
- These units need to be told what to do to the data

- An instruction processing engine consists of two components
  - Datapath: Consists of hardware elements that deal with and transform data signals
    - functional units that operate on data
    - hardware structures (e.g. wires and muxes) that enable the flow of data into the functional units and registers
    - storage units that store data (e.g., registers)
  - Control logic: Consists of hardware elements that determine control signals, i.e., signals that specify what the datapath elements should do to the data

Single-cycle vs. Multi-cycle: Control & Data

- Single-cycle machine:
  - Control signals are generated in the same clock cycle as the one during which data signals are operated on
  - Everything related to an instruction happens in one clock cycle (serialized processing)

- Multi-cycle machine:
  - Control signals needed in the next cycle can be generated in the current cycle
  - Latency of control processing can be overlapped with latency of datapath operation (more parallelism)

Flash-Forward: Performance Analysis

- Execution time of an instruction
  - {CPI} x {clock cycle time}
- Execution time of a program
  - Sum over all instructions [{CPI} x {clock cycle time}]
  - **{# of instructions} x {Average CPI} x {clock cycle time}**

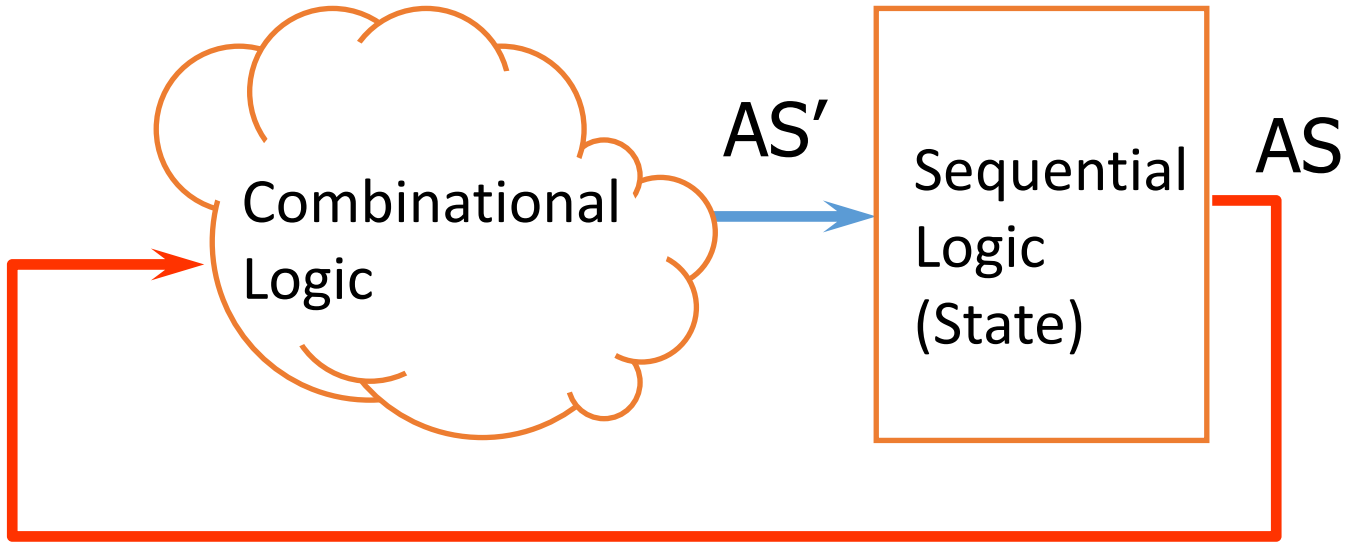
- Single-cycle microarchitecture performance
  - CPI = 1
  - Clock cycle time = long
- Multi-cycle microarchitecture performance
  - CPI = different for each instruction
    - Average CPI → hopefully small
  - Clock cycle time = short

Here, we have
two degrees of freedom
to optimize independently

A Single-Cycle Microarchitecture
A Closer Look

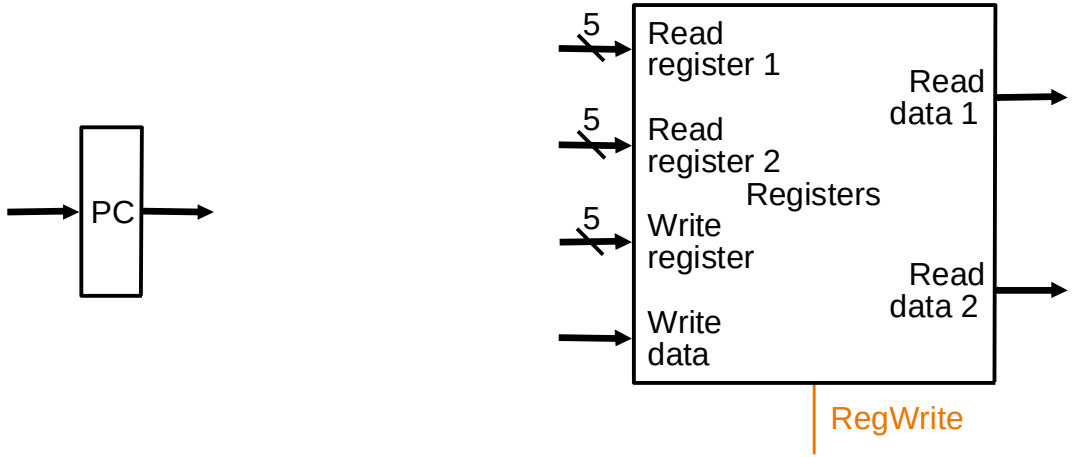
Remember…

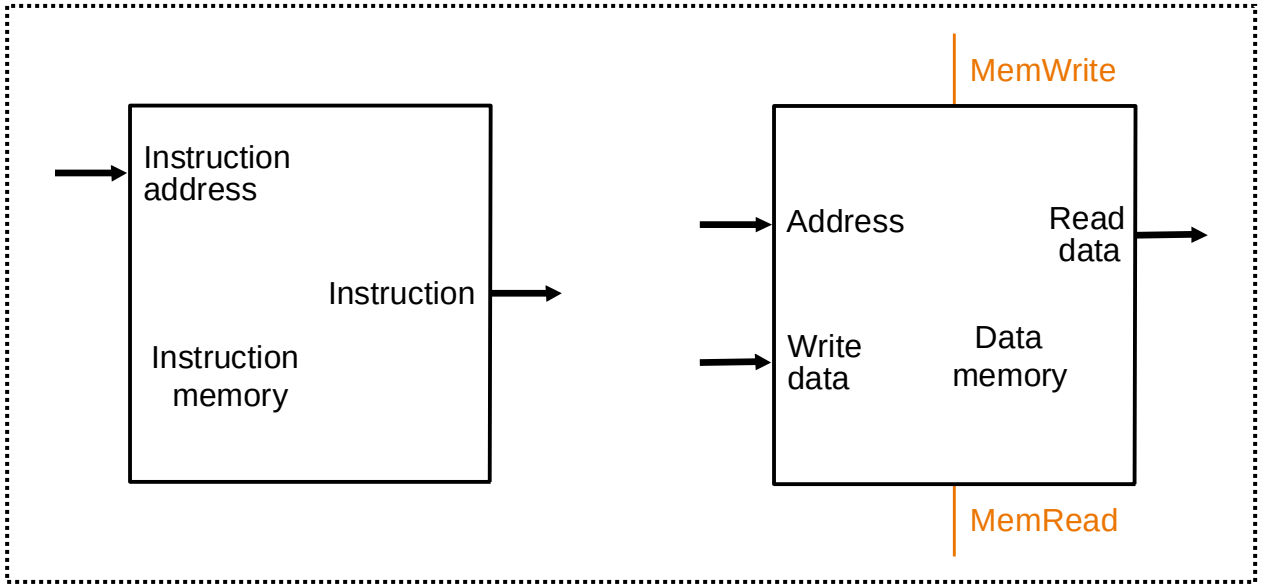
- Single-cycle machine



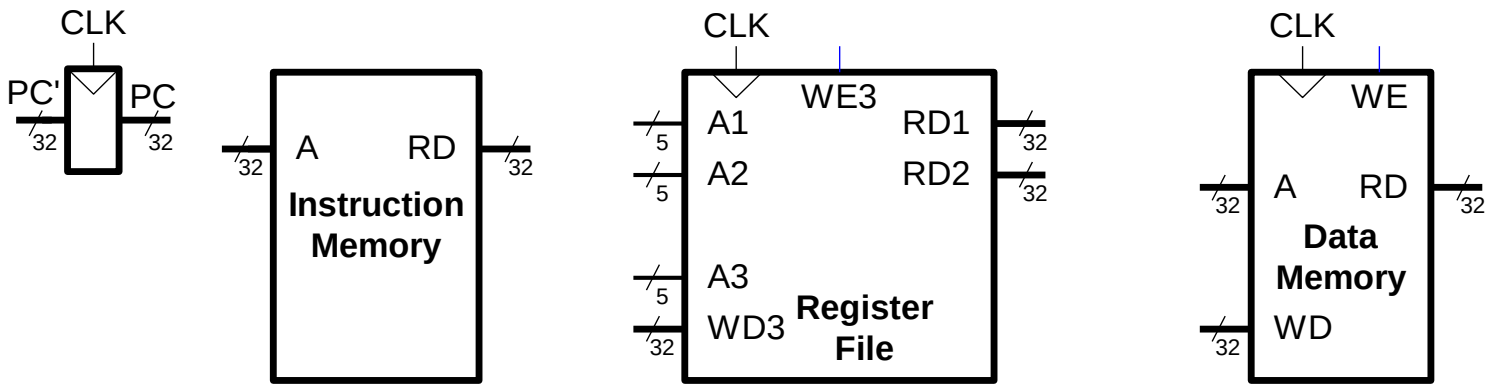
Let's Start with the State Elements

- Data and control inputs





MIPS State Elements



- Program counter:

  32-bit register

- Instruction memory:

  Takes input 32-bit address A and reads the 32-bit data (i.e., instruction) from that address to the read data output RD.

- Register file:

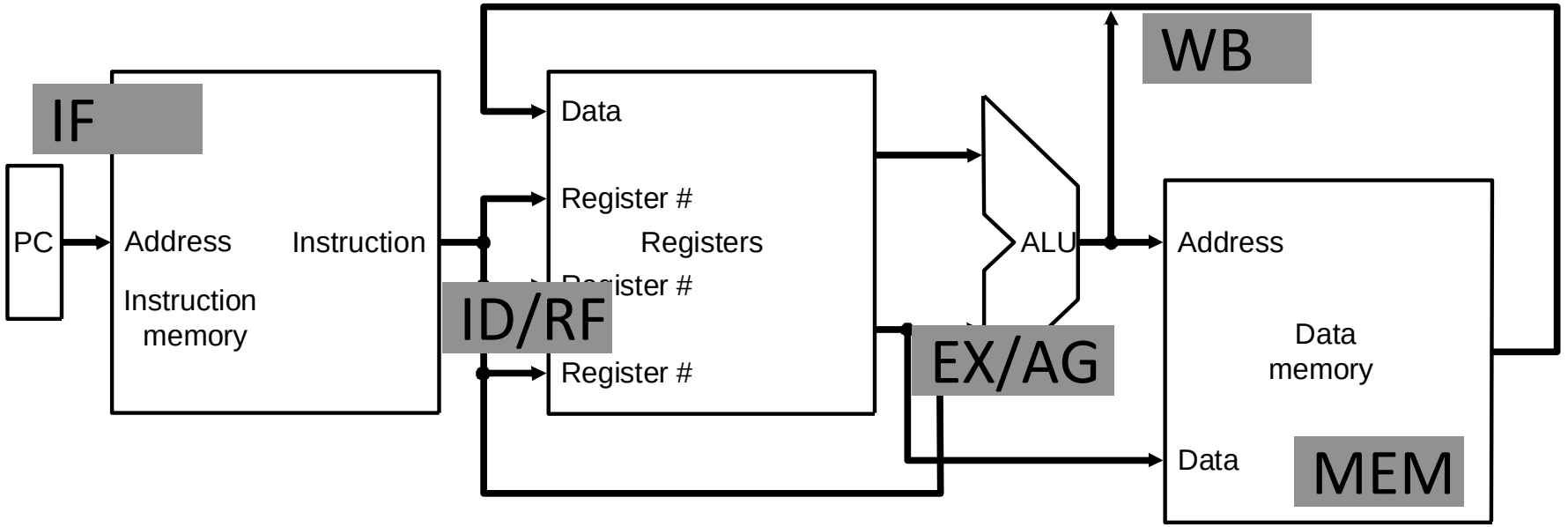
  The 32-element, 32-bit register file has 2 read ports and 1 write port

- Data memory:

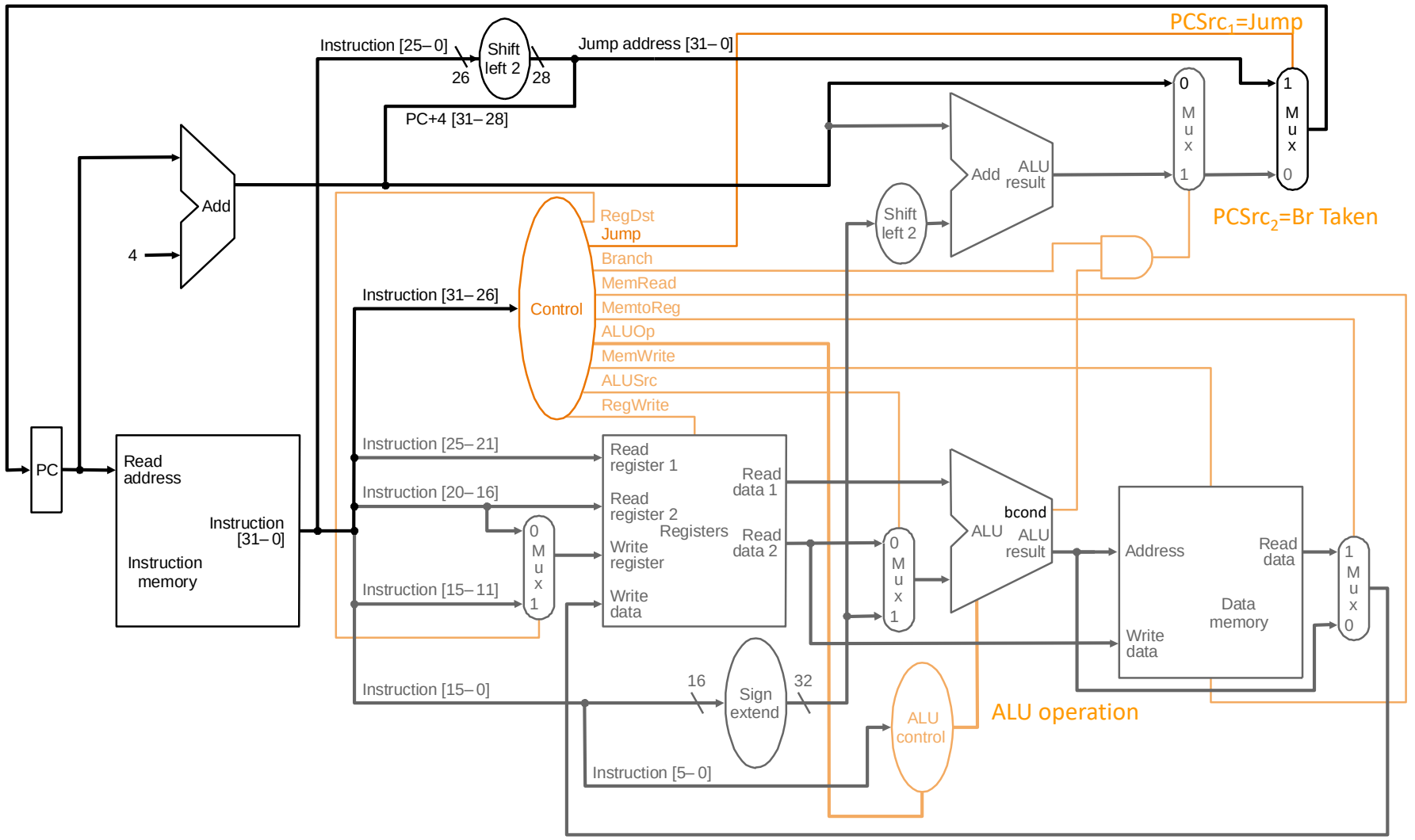
  Has a single read/write port. If the write enable, WE, is 1, it writes data WD into address A on the rising edge of the clock. If the write enable is 0, it reads address A onto RD.

Instruction Processing

- 5 generic steps
  - Instruction fetch (IF)
  - Instruction decode and register operand fetch (ID/RF)
  - Execute/Evaluate memory address (EX/AG)
  - Memory operand fetch (MEM)
  - Store/writeback result (WB)



What Is To Come: The Full MIPS Datapath



JAL, JR, JALR omitted

Another Complete Single-Cycle Processor